PRACTICE SHEET

1. What are coordinates of the point equidistant from the points (a, 0, 0), (0, a, 0), (0, 0, a) and (0, 0, 0)?

(a) $\left(\frac{a}{3},\frac{a}{3},\frac{a}{3}\right)$ (b) $\left(\frac{a}{2},\frac{a}{2},\frac{a}{2}\right)$

(c) (a, a, a) (d) (2a, 2a, 2a)

2. What is the length of the perpendicular from the origin to the plane $ax+by+\sqrt{2abz}=1$?

(a) 1 / (ab) (b) 1 / (a + b)

(c) a + b (d) ab

3. How many arbitrary constants are there in the equation of a plane?

(a) 2 (b) 3

(c) 4 (d) Any finite number

4. If P, Q are (2, 5, −7), (−3, 2, 1) respectively, then what are the direction ratios of the line PQ?

(a) <10, 6, −16> (b) <5, 3, 8>

(c) <−5, −3, −8> (d) None of these

5. If O, P are the points (0, 0, 0), (2, 3, −1) respectively, then what is the equation to the plane through P at right angles to OP?

(a) 2x + 3y + z = 16 (b) 2x + 3y − z = 14

(c) 2x + 3y + z = 14 (d) 2x + 3y − z = 0

6. Which one of following is correct?

The three planes 2x + 3y − z −2 = 0, 3x + 3y + z − 4 = 0, x − y + 2z − 5 = 0 intersect

(a) at a point (b) at two points

(c) 1 : 2 (d) 7 : 5

12. Under which one of the following conditions will the two planes x + y + z = 7 and αx + βy + γz = 3, be parallel (but not coincident)?

(a) $\alpha=\beta=\gamma=1$ only (b) $\alpha=\beta=\gamma=\frac{3}{7}$ only

(c) $\alpha=\beta=\gamma$ (d) None of the above

13. The straight line $\frac{x-3}{2}=\frac{y-4}{3}=\frac{z-5}{4}$ is parallel to which one of the following?

(a) 4x + 3y − 5z = 0 (b) 4x + 5y − 4z = 0

(c) 4x + 4y − 5z = 0 (d) 5x + 4y − 5z = 0

14. The equation $by + cz + d = 0$ represents a plane parallel to which one of the following?

(a) x − axis (b) y − axis

(c) z − axis (d) None of these

15. Which one of the following planes is normal to the plane $3x + y + z = 5$?

(a) $x + 2y + z = 6$ (b) $x - 2y + z = 6$

(c) x + 2y − z = 6 (d) x − 2y − z = 6

16. If the radius of the sphere $x^2 + y^2 + z^2 - 6x - 8y + 10z + \lambda = 0$ is unity, what is the value of λ?

(a) 49 (b) 7

(c) − 49 (d) −7

17. Curve of intersection of two spheres is

(a) an ellipse (b) a circle